

ZERO-DAY

Smart India Hackathon 2026 • Problem Statement 26145 • NTRO

Team ZERO-DAY | Dual-Layer Anomaly Detection | Read-Only Passive Defence

The Problem in 30 Seconds

- Power grids, banks & nuclear plants use hardware DATA DIODES to copy traffic into a secured monitoring enclave.
- Data flows ONE WAY only → the system can SEE everything, but can NEVER probe, re-contact, or block.
- **This unidirectional blind spot is exactly where attackers hide.**
- **The challenge: detect 6 classes of cyber threat using PURELY PASSIVE observation.**

Why Unidirectional?

✓ Air-Gap Security

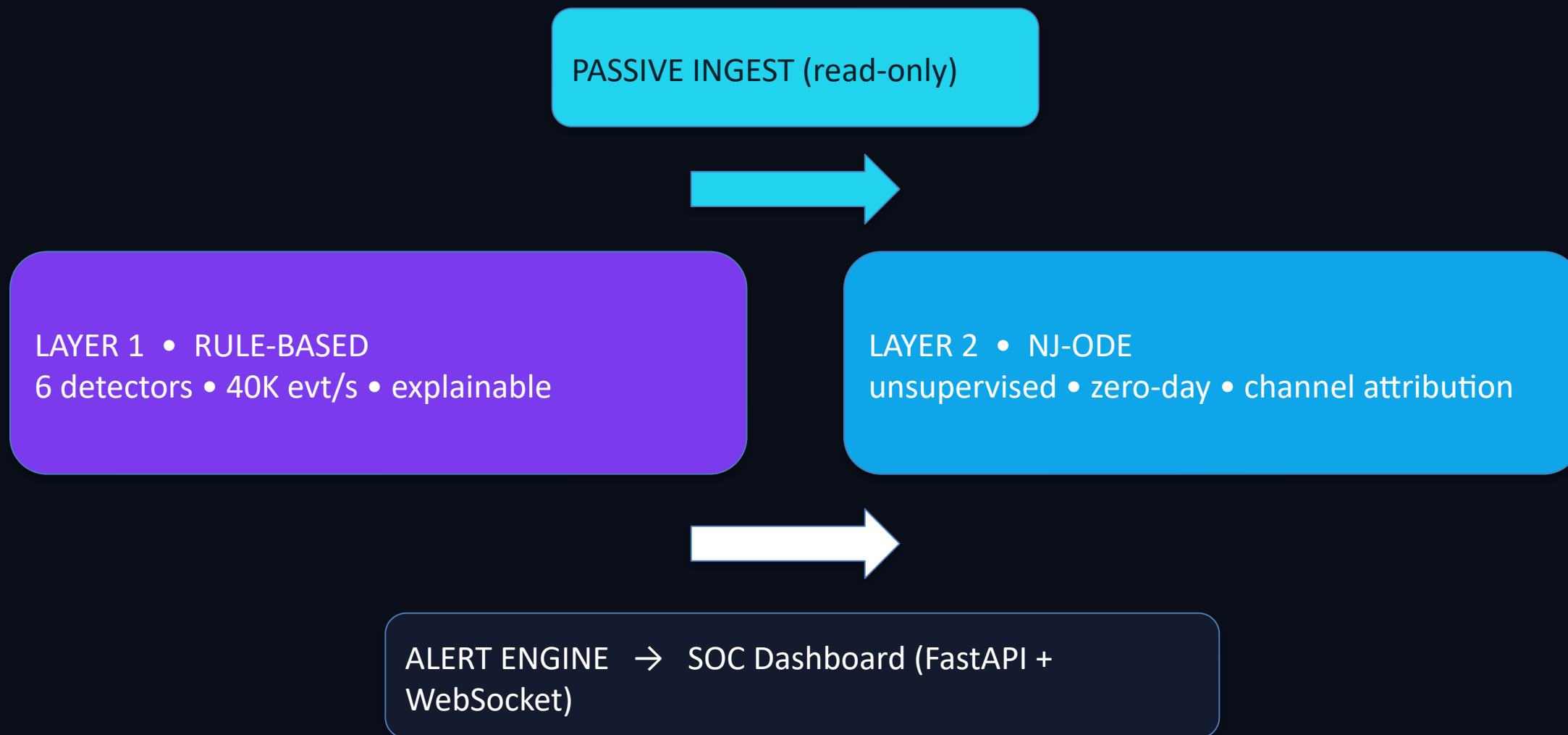
No return path = attackers on the internet can't reach the monitor.

✗ The Trade-off

We lose the ability to respond, probe, or confirm — detection must be passive.

The data diode is a one-way mirror: total visibility, zero accessibility.

Our Solution: Dual-Layer Detection



Layer 1 — Rule-Based Detectors

- Six detectors on a common interface — DDoS, C2 Beacon, DGA, DNS Tunnel, Port Scan, Exfiltration.
- Streaming: each FlowEvent processed one-at-a-time with bounded rolling-window state.
- Every alert carries evidence — WHAT (threat class), HOW confident, and WHY (feature + reason).
- **Fast & explainable: 40,202 events/sec, under 1 ms per event.**

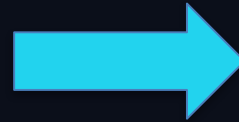
Layer 2 — Neural Jump ODE

- Unsupervised anomaly detection — trains on BENIGN traffic only, no labels needed.
- Learns the continuous-time expectation of normal traffic (Herrera, Krach & Teichmann — ICLR 2021).
- Anomaly score = prediction error between expected and observed behaviour.
- **CHANNEL ATTRIBUTION** tells analysts which feature broke (IAT, bytes, entropy, burst, direction) → maps to a threat class.

The X-Factor: Zero-Day Detection

RULES know what ATTACKS look like.

Catch what you've already seen.
Miss everything you haven't.



NJ-ODE knows what NORMAL looks like.

Anything that isn't normal gets flagged —
even attacks never seen before.

This is the whole point of AI-based detection: defence
against the unknown.

Threat Coverage — SIH 26145

- Volumetric / Protocol DDoS — SYN flood + UDP amplification.
- Botnet C2 Beaconing — periodic callbacks, detected by IAT periodicity.
- DGA Domains — malware generating random domain names via entropy scoring.
- DNS Tunnelling — data hidden inside DNS queries (length + record-type anomalies).
- Encrypted-Session Malware — TLS metadata behavior, NO decryption.
- **Reconnaissance / Port Scan & Data Exfiltration — fan-out + volumetric asymmetry.**

The Encrypted-Traffic Question

- **Hard constraint: TLS/QUIC payloads are never decrypted.**
- Instead, we analyze TLS METADATA: packet-size coefficient of variation, timing variation, JA3/JA4 fingerprints.
- **Malicious sessions have a different statistical rhythm than benign ones — even when encrypted.**

Measured Performance

40,202 evt/s

RULES-ONLY • < 1 ms per event

70 evt/s

RULES + NJ-ODE • ~14 ms per event

26 / 26 TESTS PASSING

Streaming, not batch — bounded latency

96 WINDOWS • ~5s TRAIN

NJ-ODE training in 60 epochs

Security Properties & PS Compliance

- Read-only ingest — no probes, no return path, no inline blocking.
- No payload decryption — TLS/QUIC metadata only.
- Air-gap compatible — requires no return path.
- Evidence chain — every alert carries cited feature values.
- Bounded state & thread-safe engine — deque pruning + cooldowns + N-of-M hysteresis → no alert storms.

Validation & Honest Next Steps

- Current: synthetic-vs-synthetic evaluation — 100% detection rate across all 6 attack scenarios.
- **On real data, we're clear: production validation needs CIC-IDS2017 / CSE-CIC-IDS2018.**
- That real-dataset validation is our clearly-scoped next step.

Tech Stack

- Python 3.11+ • PyTorch 2.0+ • NJ-ODE (continuous-time anomaly model)
- Pydantic v2 — strict alert schema with evidence chain
- FastAPI + WebSocket — real-time API and alert streaming
- React + TypeScript — SOC-style operations dashboard
- [pytest](#) — 26 passing tests guarding the detection engine

ZERO-DAY — Questions?

Team ZERO-DAY • SIH 2026 • PS 26145 (NTRO)

THANK YOU